Understanding Technical Indicators & Signal Assessments

A complete guide to watchlist-monitor's indicators, assessment labels, and their relationships

Who this is for: Anyone using watchlist-monitor who wants to understand what RSI, SMA, golden/death cross and the assessment labels actually mean — and crucially, what they do *not* mean.

> ■ **Important:** All indicators are descriptions of current or past conditions. None of them predict future prices. This tool is decision-support, not financial advice.

Chapter 1: The Four Indicators

1.1 RSI — Relative Strength Index (Wilder's, 14-period)

RSI measures the **speed and size of recent price changes** on a 0–100 scale. Invented by J. Welles Wilder Jr. in 1978, it is one of the most widely-used momentum indicators in the world.

How it is calculated:

Step 1 — Compute daily changes: today's close minus yesterday's close.

Step 2 — Separate gains (positive changes) from losses (absolute negative changes).

Step 3 — Seed with the simple average of the first 14 gains and 14 losses.

Step 4 — Wilder's smoothing for every day after:

```
avg_gain = (prev_avg_gain × 13 + today_gain) / 14
avg_loss = (prev_avg_loss × 13 + today_loss) / 14
```

Step 5 — RS = avg_gain / avg_loss

Step 6 — RSI = 100 – (100 / (1 + RS))

RSI Range	Zone	What it generally signals
0 – 30	Oversold	Price fell fast; some mean-reversion traders watch closely
30 – 50	Weakly bearish	Momentum tilted downward but not extreme
~50	Neutral midpoint	Roughly equal gains and losses
50 – 70	Weakly bullish	Momentum tilted upward but not extreme
70 – 100	Overbought	Price rose fast; some traders watch for pullbacks

Key nuances: RSI can stay above 70 for months (NVIDIA 2023) or below 30 through an entire bear market. It is a **momentum measure, not a forecaster.**

1.2 SMA — Simple Moving Average

An SMA averages closing prices over N trading days, smoothing out noise to reveal the underlying trend direction. Formula: SMA(n) = sum of last n closes ÷ n

SMA	Period	What it tracks	Why it matters
Short SMA	50 days	Medium-term trend (~10 weeks)	Watched by most active traders
Long SMA	200 days	Long-term trend (~10 months)	The most-watched trend divider in finance

Price above 200-day SMA = long-term bull context. **Price below 200-day SMA** = long-term bear context. Both SMAs lag — they reflect where price *has been*, not where it is going.

1.3 MA Crossover — Golden Cross & Death Cross

A crossover fires when the short SMA and long SMA change relative positions between the previous bar and the current bar.

Event	Condition	Common interpretation
Golden Cross	Short SMA crosses ABOVE long SMA	Medium-term trend overtook long-term — potential uptrend start
Death Cross	Short SMA crosses BELOW long SMA	Medium-term trend fell below long-term — potential downtrend s
None	No change in relative position	No crossover event today

Critical: Crossovers are heavily **lagging**. The S&P; 500's 2020 golden cross appeared *after* the index had already rallied 40% from its COVID lows. They confirm trends that have already begun — they do not predict them. In choppy sideways markets they generate frequent false signals.

1.4 Daily % Change

Formula: ((today_close – yesterday_close) / yesterday_close) × 100

A large single-day move (above your configured threshold, default 3%) is a flag that **something happened** — earnings, news, analyst action, sector rotation, or unusual volume. It is a **prompt to investigate**, not a trading signal by itself. The other indicators tell you the context of that move.

Chapter 2: The Four Assessment Labels

The assessment system is **fully deterministic** — no ML, no randomness, no AI. Each indicator casts a vote; the label comes from the vote tally.

Signal fired	Vote cast
RSI >= overbought threshold (default 70)	+ 1 bull
RSI <= oversold threshold (default 30)	+ 1 bear
Golden cross today	+ 1 bull
Death cross today	+ 1 bear
Large daily move	Noted in rationale — no vote

Label	Bull votes	Bear votes	Meaning
Technically strong	> 0	0	All signals bullish
Technically weak	0	> 0	All signals bearish
Looks stretched	> 0	> 0	Conflicting signals
Mixed / neutral	0	0	No signals fired today

Technically Strong

When it fires: RSI overbought and/or golden cross fired; no bearish signals.

What it means: Momentum indicators align bullishly. The stock has risen fast recently or its medium-term trend just overtook the long-term average.

Important: Does NOT mean 'will keep rising.' Overbought can persist in strong uptrends or reverse sharply. Always check fundamentals and news.

Technically Weak

When it fires: RSI oversold and/or death cross fired; no bullish signals.

What it means: Momentum indicators align bearishly. Heavy recent selling or medium-term trend has fallen below the long-term average.

Important: Does NOT mean 'will keep falling.' Oversold can mean mean-reversion opportunity OR continued decline. Context matters enormously.

Looks Stretched

When it fires: Both bullish AND bearish signals fire simultaneously.

What it means: Example: RSI overbought (bull) + death cross (bear). Or RSI oversold (bear) + golden cross (bull). The technical picture is genuinely contradictory.

Important: Often an inflection point. Shorter and longer-term timeframes disagree. Warrants the most careful investigation of all four labels.

Mixed / Neutral

When it fires: No RSI thresholds breached and no crossover today.

What it means: No notable technical condition by your configured thresholds. The most common label — most stocks on most days.

Important: Does NOT mean safe, good, or boring — just that the screener's specific conditions weren't met. Adjust thresholds in Settings if too many are neutral.

Chapter 3: How the Indicators Relate

3.1 RSI and Price Trend (SMA Relationship)

Price vs 200-day SMA	RSI	Combined picture	Assessment likely
Above (long-term uptrend)	>=70 (overbought)	Strong momentum in uptrend	Technically strong
Above (long-term uptrend)	<=30 (oversold)	Dip in uptrend — short-term weakness	Technically weak (but may reverse)
Below (long-term downtrend)	>=70 (overbought)	Sharp bounce in downtrend — dead cat?	Technically strong (high risk)
Below (long-term downtrend)	<=30 (oversold)	Downtrend with heavy selling	Technically weak — strongest bear

3.2 RSI and MA Crossovers: Timeframe Mismatch

RSI responds to the **last ~14 days**. MA crossovers reflect **weeks-to-months** of price history. They naturally disagree at market turning points — that is exactly when you get a **'Looks stretched'** label.

Scenario	RSI	MA Cross	Label	What it suggests
Bounce in downtrend	Overbought	Death cross	Looks stretched	Short-term rally within longer downtre
Trend turning up	Overbought	Golden cross	Technically strong	Momentum confirms new uptrend
Dip in uptrend	Oversold	Golden cross	Looks stretched	Short-term weakness in new uptrend
Trend turning down	Oversold	Death cross	Technically weak	Momentum confirms new downtrend

3.3 Large Daily Move in Context

Large move	RSI after	vs 200-day SMA	Context
Down	Oversold (<30)	Below	Selling into downtrend — multiple bear signals
Down	Neutral (30-70)	Above	Dip in uptrend — may be noise
Up	Overbought (>70)	Above + golden cross	Breakout with multiple confirming signals
Up	Neutral (30-70)	Below	Bounce in downtrend — investigate carefully

Chapter 4: Common Misreadings

Misreading: "RSI < 30, so the stock will bounce"	**Reality:** RSI can stay below 30 for months in a severe downtrend (many stocks in 2022). Oversold means selling pressure is high — not that a reversal is imminent.
Misreading: "Golden cross appeared — time to buy"	**Reality:** Crossovers are backward-looking. By the time a golden cross prints, the rally may be 20-40% old. It confirms a trend already in progress.
Misreading: "'Technically strong' means it's a good investment"	**Reality:** A label reflects today's technical conditions only. It says nothing about valuation, earnings quality, debt, competitive position, or macro risk.
Misreading: "'Technically weak' means I should sell"	**Reality:** 'Technically weak' in a long-term uptrend is often a buying opportunity for mean-reversion strategies. The same signal means different things depending on your timeframe and strategy.
Misreading: "More signals = more certain"	**Reality:** Multiple confirming signals increase the probability of a condition, not its certainty. All technical signals have significant failure rates.

Chapter 5: Threshold Configuration

<table>
<tr><th>Setting</th><th>Default</th><th>What it controls</th></tr>
<tr><td>daily_move_pct</td><td>3.0%</td><td>Flags tickers where |daily change| >= this</td></tr>
<tr><td>rsi_oversold</td><td>30</td><td>Bear vote + alert when RSI at or below this</td></tr>
<tr><td>rsi_overbought</td><td>70</td><td>Bull vote + alert when RSI at or above this</td></tr>
<tr><td>ma_short</td><td>50 days</td><td>Short SMA window</td></tr>
<tr><td>ma_long</td><td>200 days</td><td>Long SMA window (needs 200+ bars of data)</td></tr>
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Reset your thresholds. The test values in the DB (rsi_oversold: 99, daily_move_pct: 0.1) flag nearly everything. Open the Settings tab and set: daily_move_pct=3.0, rsi_oversold=30, rsi_overbought=70.

Appendix: Glossary

Term	Definition
Momentum	The speed of recent price movement. High RSI = high upward momentum.

Term	Definition
Mean reversion	The tendency of extreme moves to return toward average. Basis for oversold strategies.
Lagging indicator	Reflects past price history. RSI and SMA are both lagging — they do not predict.
Oversold	RSI in the 0-30 zone. Selling pressure is high. Not a guaranteed reversal.
Overbought	RSI in the 70-100 zone. Buying pressure is high. Not a guaranteed pullback.
Support	A price level where demand has historically emerged — often near the 200-day SMA.
Resistance	A price level where supply has historically appeared and slowed rallies.